

SkinCare Ecommerce Project

Professional Project Documentation

1. Project Overview

Full-stack skincare ecommerce platform built using React, Redux Toolkit, Tailwind CSS, Node.js, Express.js and MongoDB. Includes authentication, product management, cart, wishlist, coupons, notifications, AI Skin Test, referral system, admin panel and payments.

2. Technology Stack

Frontend: React, Vite, Tailwind CSS, Redux Toolkit, Axios. Backend: Node.js, Express.js, MongoDB, Mongoose. Authentication: JWT, Cookies. Email: Nodemailer. AI: Gemini API. Payments: Razorpay.

3. Authentication Module

User Registration, Login, Logout, Email Verification, Forgot Password, Reset Password, Profile Management, Change Password and Protected Routes.

4. Security Features

JWT Authentication, Password Hashing using bcrypt, Helmet, Rate Limiting, Mongo Sanitize, Cookie Based Authentication and Route Protection.

5. Product Management

Product CRUD, Product Images, Categories, Skin Types, Ingredients, Benefits, Product Variants (10ml, 50ml etc.), Related Products and Recently Viewed Products.

6. Cart & Wishlist

Persistent Cart, Quantity Management, Variant Support, Wishlist Management, Navbar Count Synchronization and Redux State Management.

7. Orders & Checkout

Checkout Flow, Address Selection, Coupon Application, Razorpay Integration, Order Placement, Order Tracking, Invoice Generation and Order Cancellation.

8. Notifications System

Order Notifications, Coupon Notifications, Read/Unread Management, Delete Notifications, Navbar Notification Badge and Real-Time Updates.

9. AI Skin Test

Gemini AI powered skin analysis. Collects user skin information and generates: Skin Summary, Morning Routine, Night Routine, Recommended Ingredients, Avoid Ingredients and Extra Tips.

10. Admin Panel

Dashboard Analytics, Product Management, Coupon Management, Order Management, Inventory Updates and Business Monitoring.

11. Referral System

Referral Codes, Referral Tracking, Reward Coupons, Referral Limits, Referral Notifications and Welcome Rewards.

12. Welcome Offer System

New users receive welcome discounts for a limited duration. Dynamic discount display across Product Cards, Product Details, Cart and Checkout.

13. Product Compare Feature

Compare multiple products, persistent storage using localStorage, compare specifications and pricing.

14. Database Collections

Users, Products, Orders, Coupons, Notifications, Reviews and Addresses.

15. Folder Structure

Frontend: components/ pages/ routes/ store/ api/ Backend: controllers/ routes/ models/ middleware/ utils/

16. Major Achievements

Implemented enterprise-level ecommerce workflows, AI integration, notifications, referrals, payments and advanced user management.

17. Future Roadmap

Deployment, Advanced Product Compare, AI Product Recommendations, Analytics Dashboard Enhancements, Sales Reports and Marketing Automation.

18. Project Evaluation

Frontend Skill Level: Advanced Intermediate Backend Skill Level: Intermediate Full Stack Capability: Strong Portfolio Project Suitable for internships and full-stack developer interviews.